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INDUSTRY TASK 1: Employee Performance & Payroll Dashboard

```
import java.awt.*;
import java.awt.event.*;

public class EmployeePayrollDashboard extends Frame implements ActionListener {

    // Input components
    private TextField txtEmployeeId;
    private TextField txtEmployeeName;
    private Choice choiceDepartment;
    private Choice choiceDesignation;
    private TextField txtBasicSalary;
    private TextField txtWorkingDays;

    private CheckboxGroup ratingGroup;
    private Checkbox excellent;
    private Checkbox good;
    private Checkbox average;
    private Checkbox poor;

    // Output
    private TextArea reportArea;

    // Buttons
    private Button btnGenerate;
    private Button btnReset;
    private Button btnExit;
```

```
public EmployeePayrollDashboard() {

    setTitle("Employee Performance & Payroll Dashboard");

    setSize(800, 650);

    setLayout(new BorderLayout(10, 10));

    // -----
    // TITLE
    // -----
    Label title = new Label(
        "EMPLOYEE PERFORMANCE & PAYROLL DASHBOARD",
        Label.CENTER
    );
    title.setFont(new Font("Arial", Font.BOLD, 20));

    add(title, BorderLayout.NORTH);

    // -----
    // MAIN PANEL
    // -----
    Panel inputPanel = new Panel();
    inputPanel.setLayout(new GridBagLayout());

    GridBagConstraints gbc = new GridBagConstraints();
    gbc.insets = new Insets(5, 5, 5, 5);
    gbc.fill = GridBagConstraints.HORIZONTAL;

    // Employee ID
    addComponent(inputPanel, new Label("Employee ID:"), gbc, 0, 0);
    txtEmployeeId = new TextField(20);
```

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```
addComponent(inputPanel, txtEmployeeId, gbc, 1, 0);

// Employee Name
addComponent(inputPanel, new Label("Employee Name:"), gbc, 0, 1);
txtEmployeeName = new TextField(20);
addComponent(inputPanel, txtEmployeeName, gbc, 1, 1);

// Department
addComponent(inputPanel, new Label("Department:"), gbc, 0, 2);

choiceDepartment = new Choice();
choiceDepartment.add("Select Department");
choiceDepartment.add("IT");
choiceDepartment.add("HR");
choiceDepartment.add("Finance");
choiceDepartment.add("Marketing");
choiceDepartment.add("Operations");

addComponent(inputPanel, choiceDepartment, gbc, 1, 2);

// Designation
addComponent(inputPanel, new Label("Designation:"), gbc, 0, 3);

choiceDesignation = new Choice();
choiceDesignation.add("Select Designation");
choiceDesignation.add("Software Engineer");
choiceDesignation.add("Senior Software Engineer");
choiceDesignation.add("Team Lead");
choiceDesignation.add("Manager");
choiceDesignation.add("HR Executive");
```

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```
addComponent(inputPanel, choiceDesignation, gbc, 1, 3);

// Basic Salary
addComponent(inputPanel, new Label("Basic Salary:"), gbc, 0, 4);
txtBasicSalary = new TextField(20);
addComponent(inputPanel, txtBasicSalary, gbc, 1, 4);

// Working Days
addComponent(inputPanel, new Label("Working Days:"), gbc, 0, 5);
txtWorkingDays = new TextField(20);
addComponent(inputPanel, txtWorkingDays, gbc, 1, 5);

// Performance Rating
addComponent(inputPanel, new Label("Performance Rating:"), gbc, 0, 6);

Panel ratingPanel = new Panel(new FlowLayout(FlowLayout.LEFT));

ratingGroup = new CheckboxGroup();

excellent = new Checkbox("Excellent", ratingGroup, false);
good = new Checkbox("Good", ratingGroup, true);
average = new Checkbox("Average", ratingGroup, false);
poor = new Checkbox("Poor", ratingGroup, false);

ratingPanel.add(excellent);
ratingPanel.add(good);
ratingPanel.add(average);
ratingPanel.add(poor);

addComponent(inputPanel, ratingPanel, gbc, 1, 6);
```

Java Programming
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```
// -----  
// BUTTON PANEL  
// -----  
Panel buttonPanel = new Panel(new FlowLayout());  
  
btnGenerate = new Button("Generate Report");  
btnReset = new Button("Reset");  
btnExit = new Button("Exit");  
  
btnGenerate.addActionListener(this);  
btnReset.addActionListener(this);  
btnExit.addActionListener(this);  
  
buttonPanel.add(btnGenerate);  
buttonPanel.add(btnReset);  
buttonPanel.add(btnExit);  
  
// -----  
// INPUT + BUTTON AREA  
// -----  
Panel centerPanel = new Panel(new BorderLayout());  
  
centerPanel.add(inputPanel, BorderLayout.CENTER);  
centerPanel.add(buttonPanel, BorderLayout.SOUTH);  
  
add(centerPanel, BorderLayout.CENTER);  
  
// -----  
// REPORT AREA  
// -----  
reportArea = new TextArea(
```

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```
        "" ,  
        12 ,  
        70 ,  
        TextArea.SCROLLBARS_VERTICAL_ONLY  
    );  
  
    reportArea.setEditable(false);  
  
    add(reportArea, BorderLayout.SOUTH);  
  
    // Window closing event  
    addWindowListener(new WindowAdapter() {  
        @Override  
        public void windowClosing(WindowEvent e) {  
            dispose();  
        }  
    });  
  
    setLocationRelativeTo(null);  
    setVisible(true);  
}  
  
// Helper method to position components  
private void addComponent(  
    Panel panel,  
    Component component,  
    GridBagConstraints gbc,  
    int x,  
    int y) {  
  
    gbc.gridx = x;
```

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```
gbc.gridy = y;

gbc.weightx = (x == 1) ? 1.0 : 0.0;


panel.add(component, gbc);
}


// -----
// EVENT HANDLING
// -----
@Override
public void actionPerformed(ActionEvent e) {

    if (e.getSource() == btnGenerate) {
        generatePayrollReport();

    } else if (e.getSource() == btnReset) {
        resetFields();

    } else if (e.getSource() == btnExit) {
        dispose();
    }
}


// -----
// PAYROLL PROCESSING
// -----
private void generatePayrollReport() {

    // Validate Employee ID
    String employeeId = txtEmployeeId.getText().trim();
```

Java Programming
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```
if (employeeId.isEmpty()) {  
    showMessage("Employee ID cannot be empty.");  
    return;  
}  
  
// Validate Employee Name  
String employeeName = txtEmployeeName.getText().trim();  
  
if (employeeName.isEmpty()) {  
    showMessage("Employee Name cannot be empty.");  
    return;  
}  
  
// Validate Department  
if (choiceDepartment.getSelectedIndex() == 0) {  
    showMessage("Please select a department.");  
    return;  
}  
  
// Validate Designation  
if (choiceDesignation.getSelectedIndex() == 0) {  
    showMessage("Please select a designation.");  
    return;  
}  
  
// Validate Basic Salary  
double basicSalary;  
  
try {  
    basicSalary = Double.parseDouble(  
        txtBasicSalary.getText().trim()
```



```
);

if (basicSalary <= 0) {
    showMessage("Basic salary must be greater than zero.");
    return;
}

} catch (NumberFormatException ex) {
    showMessage("Basic salary must be a valid numeric value.");
    return;
}

// Validate working days
int workingDays;

try {
    workingDays = Integer.parseInt(
        txtWorkingDays.getText().trim()
    );

    if (workingDays < 1 || workingDays > 30) {
        showMessage("Working days must be between 1 and 30.");
        return;
    }

} catch (NumberFormatException ex) {
    showMessage("Working days must be a valid integer.");
    return;
}

// Get performance rating
```

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```
Checkbox selectedRating = ratingGroup.getSelectedCheckbox();
```

```
if (selectedRating == null) {  
    showMessage("Please select a performance rating.");  
    return;  
}
```

```
String rating = selectedRating.getLabel();
```

```
// -----
```

```
// BONUS CALCULATION
```

```
// -----
```

```
double bonusRate;
```

```
switch (rating) {  
    case "Excellent":  
        bonusRate = 0.20;  
        break;  
  
    case "Good":  
        bonusRate = 0.10;  
        break;  
  
    case "Average":  
        bonusRate = 0.05;  
        break;  
  
    default:  
        bonusRate = 0.00;  
}
```

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```
double performanceBonus = basicSalary * bonusRate;

// Gross salary
double grossSalary = basicSalary + performanceBonus;

// Working-day adjustment
double adjustedGross =
    grossSalary * workingDays / 30.0;

// Tax
double tax = adjustedGross * 0.05;

// Net salary
double netSalary = adjustedGross - tax;

// -----
// GENERATE REPORT
// -----
StringBuilder report = new StringBuilder();

report.append("=====\n");
report.append("    EMPLOYEE PAYROLL REPORT\n");
report.append("=====\n");

report.append(String.format(
    "Employee ID    : %s\n",
    employeeId
));

report.append(String.format(
    "Employee Name   : %s\n",
```

Java Programming
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```
        employeeName

    ));

    report.append(String.format(
        "Department      : %s%n",
        choiceDepartment.getSelectedItem()
    ));

    report.append(String.format(
        "Designation      : %s%n",
        choiceDesignation.getSelectedItem()
    ));

    report.append(String.format(
        "Basic Salary      : %.2f%n",
        basicSalary
    ));

    report.append(String.format(
        "Working Days       : %d%n",
        workingDays
    ));

    report.append(String.format(
        "Performance Rating: %s%n",
        rating
    ));

    report.append("-----\n");

    report.append(String.format(
```

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```
        "Performance Bonus : %.2f%n",
        performanceBonus
    ));

    report.append(String.format(
        "Gross Salary    : %.2f%n",
        grossSalary
    ));

    report.append(String.format(
        "Adjusted Gross   : %.2f%n",
        adjustedGross
    ));

    report.append(String.format(
        "Tax (5%%)         : %.2f%n",
        tax
    ));

    report.append(String.format(
        "Net Salary        : %.2f%n",
        netSalary
    ));

    report.append("=====\n");

    reportArea.setText(report.toString());
}

// -----
// RESET
```

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```
// -----  
  
private void resetFields() {  
  
    txtEmployeeId.setText("");  
    txtEmployeeName.setText("");  
    txtBasicSalary.setText("");  
    txtWorkingDays.setText("");  
  
    choiceDepartment.select(0);  
    choiceDesignation.select(0);  
  
    ratingGroup.setSelectedCheckbox(good);  
  
    reportArea.setText("");  
}  
  
// -----  
// MESSAGE BOX  
// -----  
private void showMessage(String message) {  
  
    Dialog dialog = new Dialog(this, "Validation Message", true);  
  
    dialog.setLayout(new BorderLayout(10, 10));  
  
    Label messageLabel = new Label(message, Label.CENTER);  
  
    Button okButton = new Button("OK");  
  
    okButton.addActionListener(e -> dialog.dispose());
```

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```
        dialog.add(messageLabel, BorderLayout.CENTER);

        dialog.add(okButton, BorderLayout.SOUTH);


        dialog.setSize(400, 150);

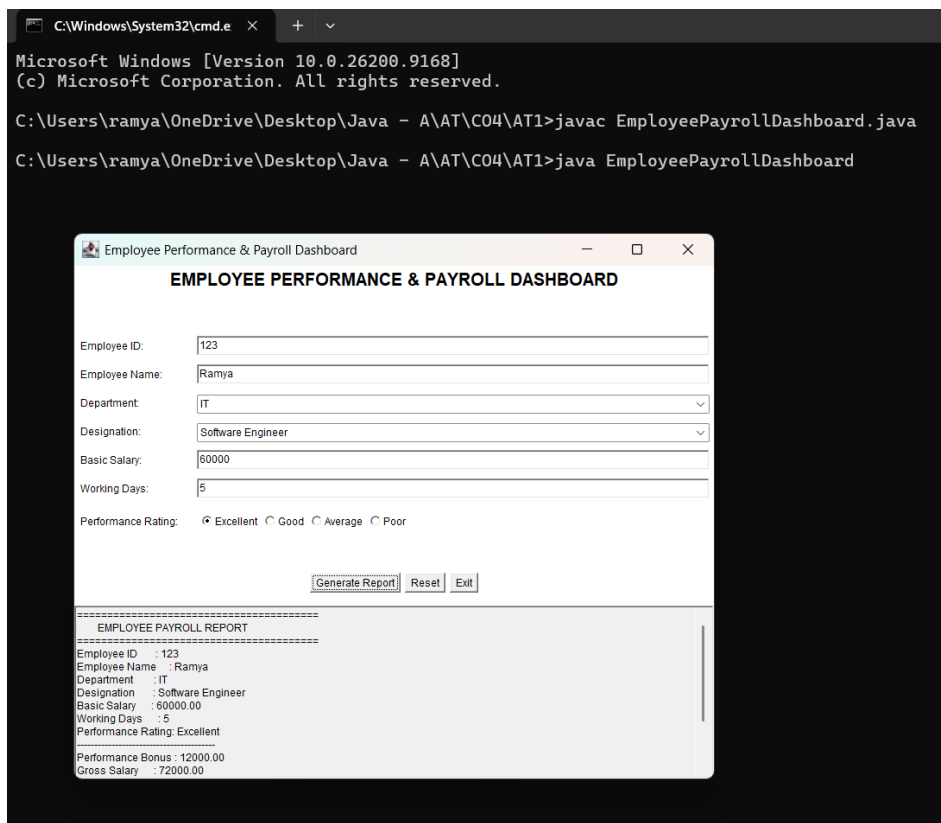
        dialog.setLocationRelativeTo(this);

        dialog.setVisible(true);
    }


    // -----
    // MAIN METHOD
    // -----

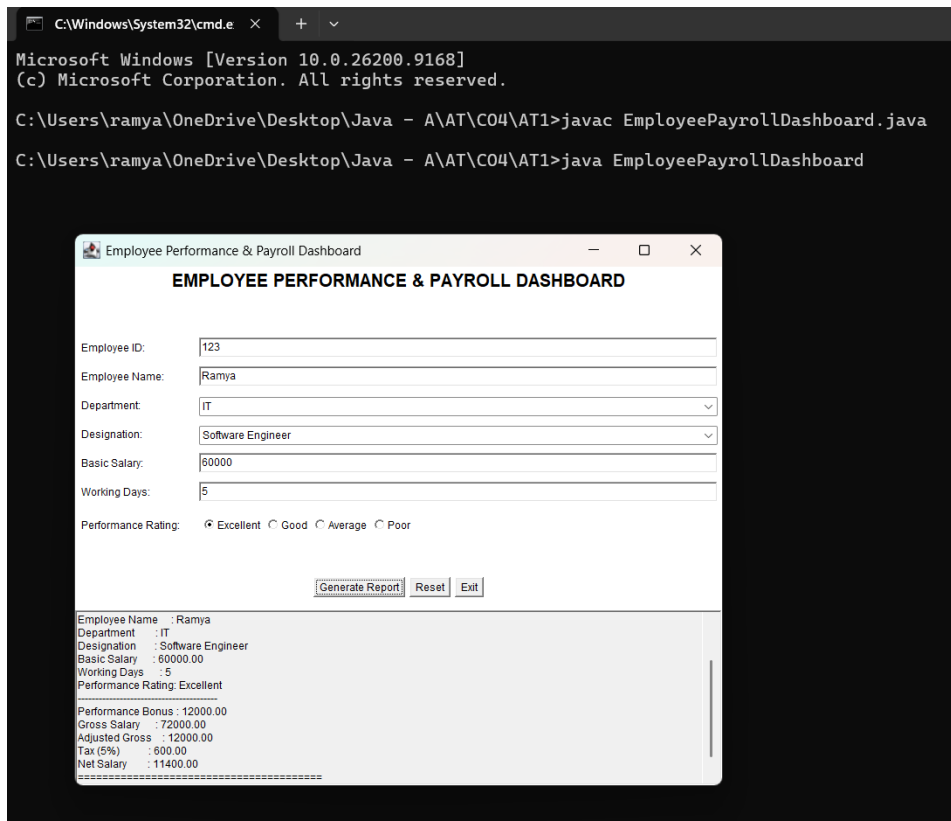
    public static void main(String[] args) {

        new EmployeePayrollDashboard();
    }
}
```



Java Programming

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INDUSTRY TASK 2: Online Food Ordering and Billing System

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
import java.util.ArrayList;
```

```
public class FoodOrderingBillingSystem
```

```
    extends Frame implements ActionListener {
```

```
    // -----
```

```
    // CARD LAYOUT
```

```
    // -----
```

```
    private CardLayout cardLayout;
```

```
    private Panel cardPanel;
```


Java Programming
CO4-AT1 : Industry Problem Solving Tasks

```
// -----  
  
// CUSTOMER DETAILS  
  
// -----  
  
private TextField txtCustomerName;  
  
private TextField txtTableNumber;  
  
  
private Button btnNext;  
  
  
  
// -----  
  
// FOOD SELECTION  
  
// -----  
  
private Choice choiceCategory;  
  
private Choice choiceFoodItem;  
  
private TextField txtQuantity;  
  
  
  
private Button btnAddItem;  
  
private Button btnGenerateBill;  
  
private Button btnPrevious;  
  
  
  
private TextArea orderArea;  
  
  
  
// -----  
  
// BILL SCREEN  
  
// -----  
  
private TextArea billArea;  
  
  
  
private Button btnClear;  
  
private Button btnBillPrevious;  
  
  
  
// -----  
  
// ORDER DATA
```

Java Programming
CO4-AT1 : Industry Problem Solving Tasks

```
// -----  
  
private ArrayList<OrderItem> orderList =  
    new ArrayList<>();  
  
// -----  
  
// FOOD DATA  
// -----  
  
private String[][] foodItems = {  
    {"Starters", "Veg Spring Roll", "120"},  
    {"Starters", "Paneer Tikka", "180"},  
    {"Starters", "Chicken 65", "220"},  
  
    {"Main Course", "Veg Fried Rice", "180"},  
    {"Main Course", "Chicken Biryani", "250"},  
    {"Main Course", "Paneer Butter Masala", "220"},  
  
    {"Beverages", "Fresh Lime Soda", "80"},  
    {"Beverages", "Cold Coffee", "120"},  
    {"Beverages", "Fresh Juice", "100"},  
  
    {"Desserts", "Gulab Jamun", "90"},  
    {"Desserts", "Ice Cream", "100"},  
    {"Desserts", "Brownie", "150"}  
};  
  
// -----  
  
// CONSTRUCTOR  
// -----  
  
public FoodOrderingBillingSystem() {  
  
    setTitle("Online Food Ordering and Billing System");
```

Java Programming
CO4-AT1 : Industry Problem Solving Tasks

```
setSize(850, 650);

// CardLayout
cardLayout = new CardLayout();
cardPanel = new Panel(cardLayout);

// Create all screens
createCustomerCard();
createFoodCard();
createBillCard();

add(cardPanel);

// Window closing
addWindowListener(new WindowAdapter() {
    @Override
    public void windowClosing(WindowEvent e) {
        dispose();
    }
});

setLocationRelativeTo(null);
setVisible(true);
}

// =====
// CARD 1 - CUSTOMER DETAILS
// =====

private void createCustomerCard() {
```

Java Programming
CO4-AT1 : Industry Problem Solving Tasks

```
Panel customerPanel = new Panel(  
    new BorderLayout(10, 10)  
);
```

```
Label title = new Label(  
    "CUSTOMER / ORDER DETAILS",  
    Label.CENTER  
);
```

```
title.setFont(  
    new Font("Arial", Font.BOLD, 22)  
);
```

```
customerPanel.add(  
    title,  
    BorderLayout.NORTH  
);
```

```
// Form
```

```
Panel formPanel = new Panel(  
    new GridLayout(5, 2, 10, 10)  
);
```

```
formPanel.add(  
    new Label("Customer Name:")  
);
```

```
txtCustomerName =  
    new TextField();
```

```
formPanel.add(  
    txtCustomerName,  
    BorderLayout.NORTH  
);
```

Java Programming
CO4-AT1 : Industry Problem Solving Tasks

```
        txtCustomerName

    );

    formPanel.add(

        new Label("Table Number:")

    );

    txtTableNumber =

        new TextField();

    formPanel.add(

        txtTableNumber

    );

    customerPanel.add(

        formPanel,

        BorderLayout.CENTER

    );

    // Next button

    Panel buttonPanel =

        new Panel(new FlowLayout());

    btnNext =

        new Button("Next");

    btnNext.addActionListener(this);

    buttonPanel.add(btnNext);

    customerPanel.add(
```

Java Programming
CO4-AT1 : Industry Problem Solving Tasks

```
        buttonPanel,

        BorderLayout.SOUTH

    );

    cardPanel.add(

        customerPanel,

        "CUSTOMER"

    );
}

// =====
// CARD 2 - FOOD SELECTION
// =====

private void createFoodCard() {

    Panel foodPanel = new Panel(

        new BorderLayout(10, 10)

    );

    Label title = new Label(

        "FOOD SELECTION",

        Label.CENTER

    );

    title.setFont(

        new Font("Arial", Font.BOLD, 22)

    );

    foodPanel.add(

        title,
```

```
        BorderLayout.NORTH

    );

    // -----
    // Selection form
    // -----
    Panel selectionPanel = new Panel(
        new GridLayout(4, 2, 10, 10)
    );

    selectionPanel.add(
        new Label("Food Category:")
    );

    choiceCategory = new Choice();

    choiceCategory.add("Select Category");
    choiceCategory.add("Starters");
    choiceCategory.add("Main Course");
    choiceCategory.add("Beverages");
    choiceCategory.add("Desserts");

    choiceCategory.addItemListener(
        e -> updateFoodItems()
    );

    selectionPanel.add(choiceCategory);

    selectionPanel.add(
        new Label("Food Item:")
    );
```

```
choiceFoodItem = new Choice();

choiceFoodItem.add(
    "Select Food Item"
);

selectionPanel.add(choiceFoodItem);

selectionPanel.add(
    new Label("Quantity:")
);

txtQuantity = new TextField();

selectionPanel.add(txtQuantity);

foodPanel.add(
    selectionPanel,
    BorderLayout.NORTH
);

// -----
// Order display
// -----
orderArea = new TextArea(
    "",
    15,
    70,
    TextArea.SCROLLBARS_VERTICAL_ONLY
);
```



```
orderArea.setEditable(false);

foodPanel.add(
    orderArea,
    BorderLayout.CENTER
);

// -----
// Buttons
// -----

Panel buttonPanel =
    new Panel(new FlowLayout());

btnPrevious =
    new Button("Previous");

btnAddItem =
    new Button("Add Item");

btnGenerateBill =
    new Button("Generate Bill");

btnPrevious.addActionListener(this);
btnAddItem.addActionListener(this);
btnGenerateBill.addActionListener(this);

buttonPanel.add(btnPrevious);
buttonPanel.add(btnAddItem);
buttonPanel.add(btnGenerateBill);
```

Java Programming
CO4-AT1 : Industry Problem Solving Tasks

```
        foodPanel.add(
            buttonPanel,
            BorderLayout.SOUTH
        );

        cardPanel.add(
            foodPanel,
            "FOOD"
        );
    }

// =====
// CARD 3 - BILL
// =====

private void createBillCard() {

    Panel billPanel = new Panel(
        new BorderLayout(10, 10)
    );

    Label title = new Label(
        "BILL / ORDER SUMMARY",
        Label.CENTER
    );

    title.setFont(
        new Font("Arial", Font.BOLD, 22)
    );

    billPanel.add(
```

Java Programming
CO4-AT1 : Industry Problem Solving Tasks

```
        title,

        BorderLayout.NORTH

    );

    billArea = new TextArea(

        "",

        20,

        70,

        TextArea.SCROLLBARS_VERTICAL_ONLY

    );

    billArea.setEditable(false);

    billPanel.add(

        billArea,

        BorderLayout.CENTER

    );

    Panel buttonPanel =

        new Panel(new FlowLayout());

    btnBillPrevious =

        new Button("Previous");

    btnClear =

        new Button("Clear");

    btnBillPrevious.addActionListener(this);

    btnClear.addActionListener(this);

    buttonPanel.add(btnBillPrevious);
```

Java Programming
CO4-AT1 : Industry Problem Solving Tasks

```
        buttonPanel.add(btnClear);

        billPanel.add(
            buttonPanel,
            BorderLayout.SOUTH
        );

        cardPanel.add(
            billPanel,
            "BILL"
        );
    }

// =====
// UPDATE FOOD ITEMS
// =====

private void updateFoodItems() {

    String selectedCategory =
        choiceCategory.getSelectedItem();

    choiceFoodItem.removeAll();

    choiceFoodItem.add(
        "Select Food Item"
    );

    for (String[] item : foodItems) {

        if (item[0].equals(selectedCategory)) {
```

```
        choiceFoodItem.add(item[1]);
    }
}

// =====
// EVENT HANDLING
// =====

@Override
public void actionPerformed(ActionEvent e) {

    if (e.getSource() == btnNext) {

        if (validateCustomerDetails()) {

            cardLayout.show(
                cardPanel,
                "FOOD"
            );
        }

    } else if (e.getSource() == btnPrevious) {

        cardLayout.show(
            cardPanel,
            "CUSTOMER"
        );

    } else if (e.getSource() == btnAddItem) {
```

```
        addItemToOrder();

    } else if (e.getSource() == btnGenerateBill) {

        generateBill();

    } else if (e.getSource() == btnBillPrevious) {

        cardLayout.show(
            cardPanel,
            "FOOD"
        );

    } else if (e.getSource() == btnClear) {

        clearOrder();
    }
}

// =====
// VALIDATE CUSTOMER DETAILS
// =====

private boolean validateCustomerDetails() {

    String customerName =
        txtCustomerName.getText().trim();

    String tableNumber =
        txtTableNumber.getText().trim();
```

```
if (customerName.isEmpty()) {  
  
    showMessage(  
        "Customer name cannot be empty."  
    );  
  
    return false;  
}
```

```
if (tableNumber.isEmpty()) {  
  
    showMessage(  
        "Table number cannot be empty."  
    );  
  
    return false;  
}
```

```
try {  
  
    int table =  
        Integer.parseInt(tableNumber);  
  
    if (table <= 0) {  
  
        showMessage(  
            "Table number must be positive."  
        );  
  
        return false;  
    }  
}
```

```
    }

    } catch (NumberFormatException ex) {

        showMessage(
            "Table number must be numeric."
        );

        return false;
    }

    return true;
}

// =====
// ADD ITEM
// =====

private void addItemToOrder() {

    // Validate category
    if (choiceCategory.getSelectedIndex() == 0) {

        showMessage(
            "Please select a food category."
        );

        return;
    }

    // Validate food item
```


Java Programming
CO4-AT1 : Industry Problem Solving Tasks

```
if (choiceFoodItem.getSelectedIndex() == 0) {

    showMessage(

        "Please select a food item."

    );

    return;
}

// Validate quantity
int quantity;

try {

    quantity =

        Integer.parseInt(

            txtQuantity.getText().trim()

        );

    if (quantity <= 0) {

        showMessage(

            "Quantity must be greater than zero."

        );

        return;
    }

} catch (NumberFormatException ex) {

    showMessage(
```

Java Programming
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```
        "Quantity must be a valid integer."
    );

    return;
}

String itemName =
    choiceFoodItem.getSelectedItem();

double price =
    getItemPrice(itemName);

if (price == -1) {

    showMessage(
        "Unable to find item price."
    );

    return;
}

// Add item to order
OrderItem orderItem =
    new OrderItem(
        itemName,
        price,
        quantity
    );

orderList.add(orderItem);
```

Java Programming
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```
        updateOrderDisplay();

        // Reset item selection
        txtQuantity.setText("");
        choiceFoodItem.select(0);
    }

    // =====
    // FIND ITEM PRICE
    // =====

    private double getItemPrice(
        String itemName) {

        for (String[] item : foodItems) {

            if (item[1].equals(itemName)) {

                return Double.parseDouble(item[2]);
            }
        }

        return -1;
    }

    // =====
    // DISPLAY CURRENT ORDER
    // =====

    private void updateOrderDisplay() {
```

Java Programming
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```
StringBuilder order =  
    new StringBuilder();  
  
order.append(  
    "CURRENT ORDER\n"  
);  
  
order.append(  
    "-----\n"  
);  
  
order.append(  
    String.format(  
        "%-25s %-10s %-10s %-10s\n",  
        "Item",  
        "Price",  
        "Qty",  
        "Cost"  
    )  
);  
  
order.append(  
    "-----\n"  
);  
  
double subtotal = 0;  
  
for (OrderItem item : orderList) {  
  
    double cost =  
        item.price * item.quantity;
```

```
        subtotal += cost;

        order.append(
            String.format(
                "%-25s %-10.2f %-10d %-10.2f%n",
                item.name,
                item.price,
                item.quantity,
                cost
            )
        );
    }

    order.append(
        "-----\n"
    );

    order.append(
        String.format(
            "Subtotal: %.2f%n",
            subtotal
        )
    );

    orderArea.setText(
        order.toString()
    );
}

// =====
```

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```
// GENERATE BILL

// =====

private void generateBill() {

    if (orderList.isEmpty()) {

        showMessage(

            "Please add at least one item before generating the bill."

        );

        return;
    }

    double subtotal = 0;

    for (OrderItem item : orderList) {

        subtotal +=

            item.price * item.quantity;
    }

    // GST = 5%

    double gst =

        subtotal * 0.05;

    double finalAmount =

        subtotal + gst;

    StringBuilder bill =

        new StringBuilder();
```

```
bill.append(  
    "=====\\n"  
);
```

```
bill.append(  
    "    RESTAURANT BILL\\n"  
);
```

```
bill.append(  
    "=====\\n"  
);
```

```
bill.append(  
    "Customer Name : "  
);
```

```
bill.append(  
    txtCustomerName.getText().trim()  
);
```

```
bill.append("\\n");
```

```
bill.append(  
    "Table Number : "  
);
```

```
bill.append(  
    txtTableNumber.getText().trim()  
);
```

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```
bill.append("\n");
```

```
bill.append(  
    "-----\n"  
);
```

```
bill.append(  
    String.format(  
        "%-25s %-8s %-8s %-10s\n",  
        "Item",  
        "Price",  
        "Qty",  
        "Amount"  
    )  
);
```

```
bill.append(  
    "-----\n"  
);
```

```
for (OrderItem item : orderList) {
```

```
    double amount =  
        item.price * item.quantity;
```

```
bill.append(  
    String.format(  
        "%-25s %-8.2f %-8d %-10.2f\n",  
        item.name,  
        item.price,  
        item.quantity,  
        amount
```


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```
        amount
    )
    );
}

bill.append(
    "-----\n"
);

bill.append(
    String.format(
        "Subtotal    : %.2f%n",
        subtotal
    )
);

bill.append(
    String.format(
        "GST (5%%)     : %.2f%n",
        gst
    )
);

bill.append(
    String.format(
        "Final Bill    : %.2f%n",
        finalAmount
    )
);

bill.append(
```

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```
        "=====\\n"

    );

    bill.append(

        "    Thank you! Visit again.\\n"

    );

    bill.append(

        "=====\\n"

    );

    billArea.setText(

        bill.toString()

    );

    // Show bill card
    cardLayout.show(

        cardPanel,

        "BILL"

    );
}

// =====
// CLEAR ORDER
// =====

private void clearOrder() {

    orderList.clear();

    txtCustomerName.setText("");
```

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```
txtTableNumber.setText("");
txtQuantity.setText("");

choiceCategory.select(0);

choiceFoodItem.removeAll();

choiceFoodItem.add(
    "Select Food Item"
);

orderArea.setText("");
billArea.setText("");

cardLayout.show(
    cardPanel,
    "CUSTOMER"
);
}

// =====
// MESSAGE BOX
// =====

private void showMessage(
    String message) {

    Dialog dialog =
        new Dialog(
            this,
            "Message",
```

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```
        true

    );

    dialog.setLayout(
        new BorderLayout(10, 10)
    );

    Label messageLabel =
        new Label(
            message,
            Label.CENTER
        );

    Button ok =
        new Button("OK");

    ok.addActionListener(
        e -> dialog.dispose()
    );

    dialog.add(
        messageLabel,
        BorderLayout.CENTER
    );

    dialog.add(
        ok,
        BorderLayout.SOUTH
    );

    dialog.setSize(450, 150);
```

```
        dialog.setLocationRelativeTo(this);

        dialog.setVisible(true);
    }

// =====
// ORDER ITEM CLASS
// =====

private static class OrderItem {

    String name;
    double price;
    int quantity;

    OrderItem(
        String name,
        double price,
        int quantity) {

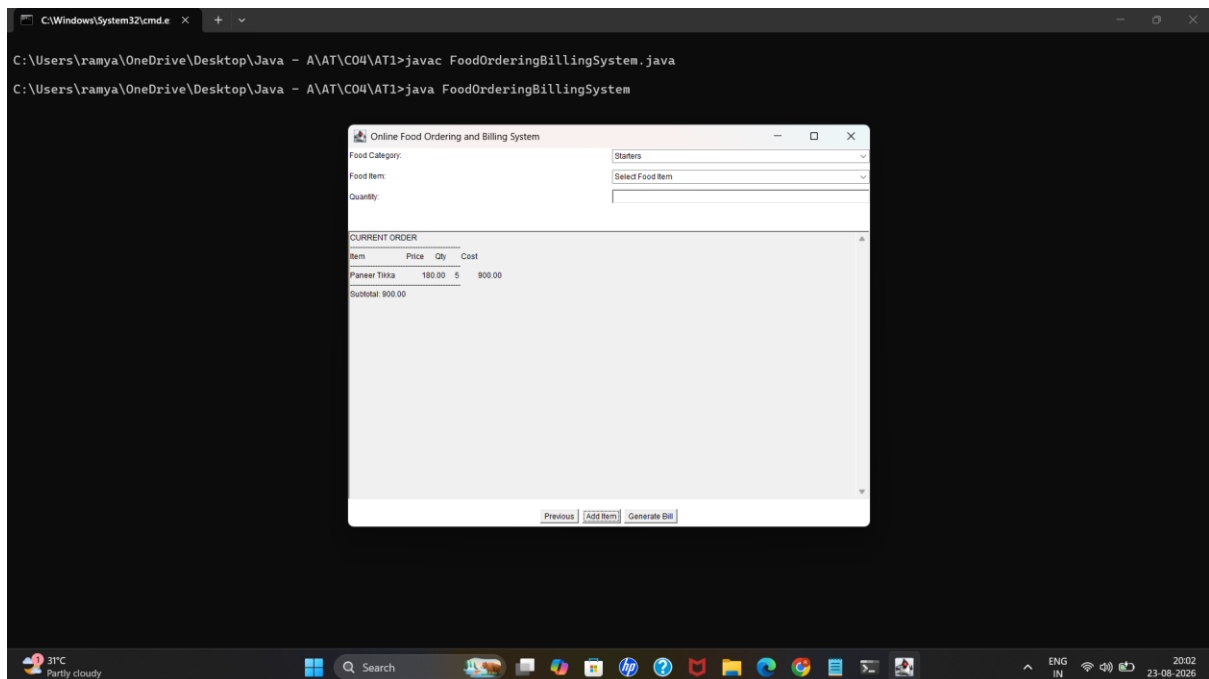
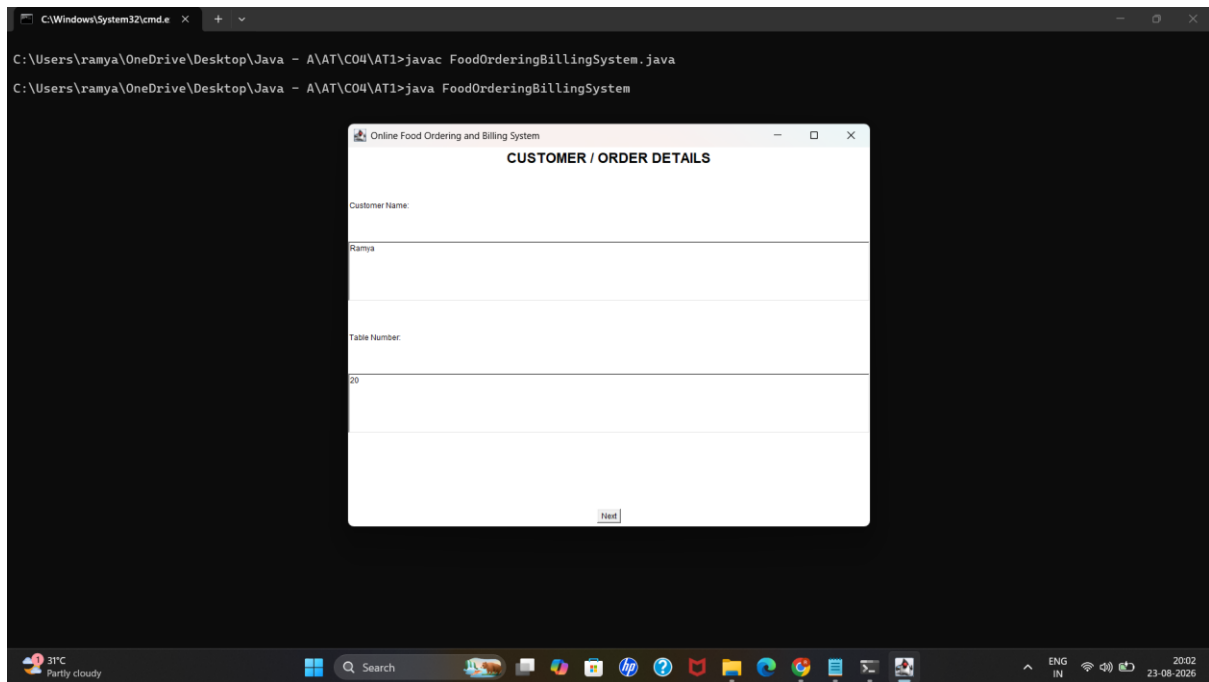
        this.name = name;
        this.price = price;
        this.quantity = quantity;
    }
}

// =====
// MAIN METHOD
// =====
```

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```
public static void main(String[] args) {  
  
    new FoodOrderingBillingSystem();  
  
}  
}
```



Java Programming

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